

Multi-Teacher Pseudo-Labeling for Document-Level Sentiment Classification

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Abstract

Most research on sentiment analysis targets short texts such as tweets or single-sentence movie review snippets. Long-form reviews are more challenging as they combine plot descriptions, mixed sentiment and rhetorical devices. In this project I explore a semi-supervised approach to document-level sentiment classification in a domain with no human-labeled data. Four transformer sentiment classifiers are used as teachers to assign pseudo-labels to TMDB movie reviews. High-confidence samples are selected via agreement and probability thresholding and used to fine-tune a compact student model (DistilBERT). Experiments show the student outperforms classical tf-idf baselines on accuracy, balanced accuracy and macro-F1 while being much cheaper than the teacher ensemble. The analysis highlights limitations of pseudo-labeling and outlines future directions.

1 Introduction

Sentiment analysis aims to automatically determine whether an opinion expressed in text is positive or negative. While the task is well-established, most benchmark datasets such as SST-2 or IMDB consist of short, well-structured inputs. In contrast, long-form reviews contain a mixture of narrative plot summary, objective description and subjective evaluation. Global polarity decisions on such documents are more difficult because sentiment is distributed across multiple paragraphs and may shift throughout the review.

In the domain considered here—movie reviews collected from TMDB—no labeled training data exist. Annotating thousands of reviews manually would be expensive and slow, and is unrealistic for a small course project. Semi-supervised approaches are therefore attractive. I adopt a *multi-teacher pseudo-labeling* strategy: an ensemble of strong transformer sentiment classifiers is used to label unlabeled reviews, and a small student model is trained on a filtered subset of high-confidence pseudo-labels.

The project investigates three questions: (1) How consistent are teacher predictions on long reviews? (2) How does confidence thresholding affect pseudo-labeled dataset quality and size? (3) Can a compact student outperform classical baselines without access to human labels?

2 Related Work and SOTA

Early sentiment analysis used bag-of-words features and linear classifiers such as logistic regression and SVMs [PangLee2008,Liu2012]. These models were simple and robust, but relied heavily on explicit sentiment words and could not easily model negation scope or long-distance dependencies.

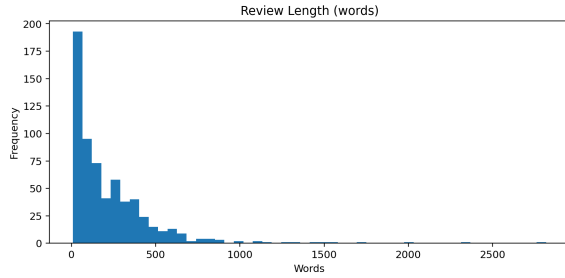
Neural models based on CNNs and LSTMs improved compositional representation, but the major leap came with transformer-based pre-trained language models. BERT [Devlin2019] introduced bidirectional pretraining with masked language modeling and next sentence prediction. RoBERTa [Liu2019RoBERTa] optimized pretraining by removing the next sentence objective, training longer and using larger mini-batches. More recent architectures such as DeBERTaV3 [He2021DeBERTaV3] incorporate disentangled attention and ELECTRA-style pretraining, achieving or approaching state-of-the-art results on GLUE and SST-2 sentiment benchmarks. DistilBERT [Sanh2019DistilBERT] distills BERT into a lighter model with around 40% fewer parameters while retaining most of its performance, making it attractive for deployment and as a student model.

Semi-supervised learning approaches for NLP include self-training and pseudo-labeling, where a model trained on labeled data generates labels for unlabeled instances. Knowledge distillation, originally proposed by Hinton and colleagues, transfers knowledge from a large teacher network to a smaller student by matching soft output distributions. Recent work such as Noisy Student and FixMatch combines pseudo-labeling with consistency regularization and confidence-based filtering. In this project, RoBERTa-large and DeBERTaV3-base sentiment models are used as near-SOTA teachers, while a DistilBERT student offers a favorable cost-performance trade-off.

3 Dataset and Task

The task is binary sentiment classification (*positive / negative*) at document level. Reviews were collected from TMDB using the public API. For each review I keep the textual content and some metadata (movie id, title, author, creation time). After filtering non-English reviews and removing very short or malformed entries, the dataset consists largely of multi-paragraph reviews.

The length distribution of the cleaned dataset is shown below and confirms the document-level nature of the task:



Since no human gold labels exist, pseudo-labeling is used to create a labeled dataset. After the labeling and filtering steps described in the next section, the final dataset is split 80/10/10 into train/validation/test using stratification by label. The resulting statistics are summarized below:

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TABLE 1 (Dataset Statistics) - I'll just copy this into my LaTeX report later
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#Samples  Positive (%)  Negative (%)
Train      459.0         87.1        12.9
Validation 82.0          86.6        13.4
Test       96.0          87.5        12.5
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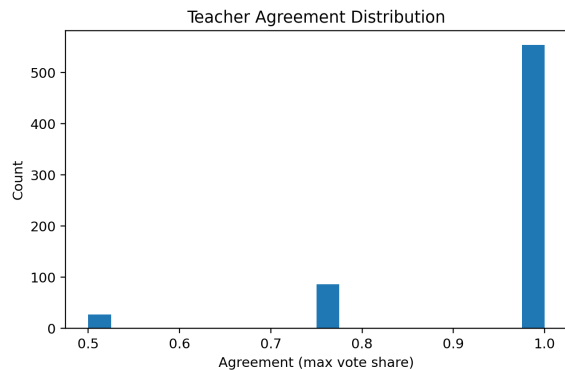
4 Methodology

4.1 Teacher Ensemble

Four transformer sentiment classifiers serve as teachers:

- siebert/sentiment-roberta-large-english
- distilbert-base-uncased/-finetuned-sst-2-english
- textattack/roberta-base-SST-2
- vroomhf/deberta-v3-base-finetuned-sst2

Each teacher outputs a probability distribution over sentiment labels for a given review. The ensemble exploits diversity across backbones (RoBERTa-large, RoBERTa-base, DistilBERT, DeBERTaV3-base) while keeping all models within the same language and task family (English SST-2-style sentiment). Teacher agreement on polarity is an important signal for label reliability; the histogram below shows how many teachers agree on the predicted label for each review:



Most reviews receive agreement from at least three teachers, which suggests that the ensemble is relatively consistent on this domain.

4.2 Confidence Thresholding

Let $p_t(y | x)$ denote the teacher t softmax confidence for label y on review x . Each teacher first predicts

$$y_t = \arg \max_y p_t(y | x).$$

Majority voting assigns the ensemble label

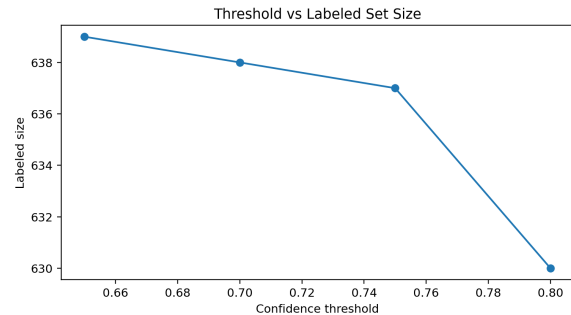
$$\hat{y} = \text{mode}(y_1, \dots, y_T)$$

where $T = 4$ is the number of teachers. Define $\mathcal{A} = \{t : y_t = \hat{y}\}$ as the set of teachers that agree with the majority. Ensemble confidence is computed as the mean confidence of agreeing teachers:

$$c(x) = \frac{1}{|\mathcal{A}|} \sum_{t \in \mathcal{A}} p_t(\hat{y} | x).$$

A review is kept if and only if $|\mathcal{A}| \geq 3$ and $c(x) \geq \tau$, where τ is a user-chosen threshold.

The threshold controls a trade-off between pseudo-label quality and dataset size. Small τ values include more examples but risk propagating noisy labels; large values reduce noise but may discard rare or difficult cases. The plot below shows how the total size of the pseudo-labeled set varies with τ :

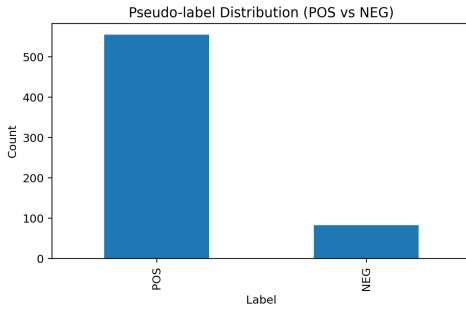


The next plot shows positive vs. negative counts for the same thresholds:

	threshold	size	POS	NEG
0	0.65	639	556	83
1	0.70	638	556	82
2	0.75	637	555	82
3	0.80	630	549	81

In this project I choose a relatively strict threshold where the ensemble is highly confident but the dataset remains large enough to train a neural model.

The final class distribution after thresholding is shown below:



Positive reviews clearly dominate, a typical pattern for movie reviews. To better understand the semantics of each class, I created word clouds:



Positive reviews emphasize words like *love*, *great*, *performance*, whereas negative ones contain terms such as *boring*, *bad*, *waste*.

4.3 Student Model

The student is DistilBERT with a linear classification head on top of the [CLS] representation. All layers are fine-tuned end-to-end using class-weighted cross-entropy to mitigate label imbalance. Optimization uses AdamW with learning rate $2 \cdot 10^{-5}$, batch size 16 and up to 3 epochs with early stopping based on validation accuracy. Compared to the teacher ensemble, the student is considerably cheaper at inference time and can be deployed on modest hardware.

5 Experiments

5.1 Baselines and Setup

As classical baselines I train logistic regression and linear SVM with tf-idf features over uni- and bigrams. These models are simple yet competitive on many sentiment tasks and provide a useful point of comparison.

All models are trained on the same pseudo-labeled training set and tuned using the validation set. Evaluation is performed on the held-out test set. Metrics include accuracy, balanced accuracy and macro-F1. The last two are important because of the skew towards positive reviews: a classifier that always predicts positive would achieve high accuracy but poor balanced accuracy and macro-F1.

5.2 Results

The plot below compares baselines and the student model on the pseudo-labeled test set:

	model	ACC	BACC	MacroF1
0	TFIDF+LogReg	0.895833	0.619048	0.659091
1	TFIDF+LinearSVM	0.875000	0.500000	0.466667

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The DistilBERT student clearly improves over tf-idf baselines in balanced accuracy and macro-F1, indicating better performance on the minority class. Overall accuracy also increases, suggesting that contextual representations are beneficial even with noisy supervision.

More detailed class-wise metrics are shown below:

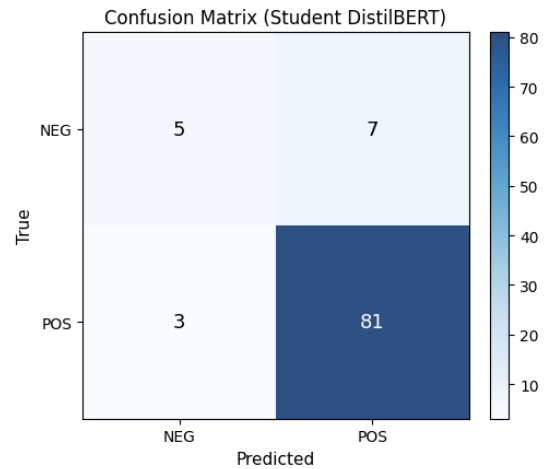
Confusion Matrix (Table Form):				
	Pred_NEG	Pred_POS		
True_NEG	5	7		
True_POS	3	81		

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	precision	recall	f1-score	support
NEG	0.625000	0.416667	0.500000	12.000000
POS	0.920455	0.964286	0.941860	84.000000
accuracy	0.895833	0.895833	0.895833	0.895833
macro avg	0.772727	0.690476	0.720938	96.000000
weighted avg	0.883523	0.895833	0.886628	96.000000

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Finally, the confusion matrix of the student model highlights where errors are concentrated:



Most mistakes occur on the negative class, which is both smaller and more diverse linguistically.

6 Discussion

Qualitative inspection of misclassified examples reveals several recurring patterns. Some reviews are genuinely mixed: the author lists both strengths and weaknesses of the movie, making the global label subjective even for humans. Other problematic cases involve sarcasm or irony, where surface-level positive words are used to express negative sentiment. A third group consists of reviews that spend many sentences on neutral plot description and only briefly state an opinion at the end.

An additional complication is that evaluation relies on pseudo-labels rather than human gold. This means that disagreements between the student and ensemble are treated as errors even if the student prediction might be closer to what a human annotator would decide. A small manually labeled set would be extremely valuable for calibration and for a more realistic estimate of absolute performance.

7 Conclusion and Future Work

Multi-teacher pseudo-labeling provides a practical way to train a document-level sentiment classifier in the absence of labeled data. In this project a compact DistilBERT student trained on high-confidence pseudo-labels outperforms classical baselines while being far cheaper than the teacher ensemble. The approach is simple to implement and could be reused for other domains with unlabeled reviews.

Future work includes collecting a small human-labeled evaluation set, extending the label space to neutral and mixed sentiment, exploring soft-label distillation from the ensemble, and using low-confidence examples in a consistency-regularized semi-supervised objective instead of discarding them.

8 Contribution

This project was conducted individually.

9 References

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